Have Molecular Foundation Models Improved for Free-Energy Sampling?

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Abstract

Foundation models offer broadly applicable potential energy surfaces that can either be applied in a zero-shot manner (without domain-specific retraining) or fine-tuned to the task at hand. Their accuracy is typically judged by energy and force errors on near-equilibrium data, yet it remains largely unclear whether these metrics translate into reliable free-energy landscapes. Here, enhanced sampling can be a particularly demanding test: simulations are driven far from equilibrium into sparsely trained regions and held there over long trajectories, so small systematic errors in forces or relative energies can accumulate and distort predicted populations, transition pathways, and barriers. We assess several molecular foundation models, spanning different generations and architectures (including models with explicit long-range electrostatics) in prototypical metadynamics and umbrella sampling simulations. Our results show that recent foundation models can reproduce simple gas-phase free-energy profiles with near-reference accuracy, but their reliability deteriorates for more complex multidimensional and solvated free-energy landscapes. While current molecular foundation models thus provide useful potentials for free-energy sampling, they are not yet universally reliable in the far-from-equilibrium regions explored by enhanced sampling, so that task-specific validation and finetuning remain warranted.

Introduction

For nearly half a century, computational chemistry has been characterized by the dichotomy between electronic structure methods (such as density functional theory - DFT), and classical force fields (FFs). Both come with their respective advantages and drawbacks: DFT is computationally demanding but can be applied across chemical space, because it is rigorously based on first-principles. In contrast, FFs are fast but restricted to specific domains and, due to their empirical nature, unable to extrapolate. With the introduction of local-environment descriptors, interatomic potentials based on machine learning (ML) have been able to provide a serious alternative to these paradigms.[1,2] Machine-learning interatomic potentials (MLIPs) combine the accuracy of the underlying reference method (usually based on first-principles) with the computational efficiency of classical FFs, enabling nanosecond-timescale simulations of thousands of atoms at electronic structure accuracy.

Originally, predefined atomic environment descriptors such as symmetry functions[1] or the Smooth Overlap of Atomic Positions (SOAP)[2] were used in MLIPs. Starting around 2017, a new generation of ML models based on graph neural networks rose in popularity.[3–7] Here, an optimized atomic environment representation is learned during the training process. To this end, the elemental species of an atom is first embedded into a vector space, which is subsequently updated according to the atomic environment using a message passing algorithm. These design choices yield models that are scalable to large datasets and across the periodic table. Modern incarnations of this approach typically include directional information in the message passing procedure. This can for example be achieved via a generalization of body-ordered local-environment descriptors such as the Atomic Cluster Expansion (ACE)[8], yielding equivariant MLIPs such as Allegro[9], NequIP[10], MACE[11], or eSEN[12]. Combined, these advances make it possible to train such models on large datasets covering diverse chemical environments, creating so-called universal or foundational MLIPs (fMLIPs).

The first generation of fMLIPs was trained on the MPTraj dataset containing near equi-

librium configurations of inorganic crystals.[13–15] While the first models already showed remarkable extrapolative capabilities (considering the limited scope of the training data), the development of larger and more diverse datasets[16–18] has enabled their continuous improvement, so that current foundation models approach the accuracy of electronic-structure methods for a wide range of simulation tasks.[19–23] This is particularly true for models such as AIMNet2[24], MACE-OFF[25], or PET-SPICE[21], which focus (bio-) organic molecular systems. These models have been shown to be highly accurate for geometry optimization, conformer ranking, thermochemistry and other tasks. Augmentation with long-range electrostatic interactions has further allowed such models to reliably describe the equilibrium properties of molecular crystals and liquids, as demonstrated for example in MACELES-OFF[26,27], MACE-POLAR[28], and SO3LR[29].

While computing special points or paths on the potential energy surface is the backbone of many molecular simulation tasks, such approaches do not fully capture the thermal and entropic contributions to the thermodynamics or kinetics of chemical processes.[30,31] These effects can naturally be captured via Molecular Dynamics (MD) simulations. However, many interesting chemical processes (in particular chemical reactions) occur on timescales that are far too slow to be captured in conventional MD simulations. In these cases, more advanced free energy simulation techniques such as metadynamics[32,33] or umbrella sampling[34,35] are used. These enable the rigorous prediction of reaction free energies and free energy barriers, by biasing the simulation along the slow degrees of freedom, which must be represented via a suitable set of collective variables (CVs).[36–41] Owing to the limited time and length scales accessible to electronic-structure methods, their application to free-energy sampling has always been restricted, however. This has left classical FFs as the most common alternative for free energy simulations on complex systems, although this comes with a significant loss in accuracy. This is a clear opportunity for MLIPs, and indeed bespoke or fine-tuned MLIPs have been used to perform free-energy sampling for challenging systems, where electronic structure methods are prohibitively expensive and classical FFs lack accuracy.[CITATION

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The high accuracy of current foundation models in typical benchmarks makes them interesting candidates for this domain. However, good performance on common potential-energy surface minimizations and scans does not automatically translate to reliable free-energy surfaces. After all, the training structures of foundation models are still mostly obtained from near-equilibrium sampling using methods such as molecular dynamics, whereas free-energy sampling drives the simulation away from equilibrium.[42] In addition to this, relatively long simulation times, often on the order of nanoseconds, are spent in these regions.[36–38] This prolonged exposure makes potential-energy drifts, unphysical structures, and holes in the PES far more likely to surface.citeStockerHowRobust

The fact that free-energy sampling stresses models in regimes where even the largest training sets are sparse, makes it an excellent test case for the extrapolative capabilities of foundation models. This paper therefore investigate how well (bio-)organic foundation models from different generations perform on representative free-energy sampling problems. In this context, we also introduce a universal biasing tool for the Atomic Simulation Environment (ASE), called biASE. This is an autograd-based engine that biases the PES along arbitrary collective variable (CV), automatically computing the required derivatives. This enables the effortless connection of fMLIP calculators to enhanced sampling methods, making the present benchmarks easily reproducible with new models. In the following, we briefly review the theoretical foundations of these techniques, and discuss their implementation in biASE. Subsequently, we motivate the choice of the fMLIPs and test cases considered herein and discuss the results of the study.

Theory

Metadynamics

Metadynamics accelerates sampling of the PES by introducing a history-dependent bias potential in the space of a set of collective variables (CVs), $\boldsymbol{\xi} = c(\mathbf{R})$.[32] For a single CV, ξ, the bias is constructed by periodically depositing Gaussian functions centered at the values visited by the system,

$$V_{\text{bias}}(\xi, t) = \sum_{t_k<t} w \exp\left[-\frac{(\xi - \xi(t_k))^2}{2\sigma^2}\right], \tag{1}$$

where w is the Gaussian height, σ is its width, and $\xi(t_k)$ denotes the value of the CV at deposition time t_k. By progressively penalizing regions that have already been sampled, the bias facilitates escape from metastable free-energy minima and promotes transitions across otherwise rarely crossed barriers.[32]

For n_c collective variables, $\boldsymbol{\xi} \in \mathbb{R}^{n_c}$, the bias generalizes to multidimensional Gaussian kernels,

$$V_{\text{bias}}(\boldsymbol{\xi}, t) = \sum_{t_k<t} w \exp\left[-\frac{1}{2}\left(\boldsymbol{\xi} - \boldsymbol{\xi}(t_k)\right)^{\mathrm{T}} \boldsymbol{\Sigma}^{-1} \left(\boldsymbol{\xi} - \boldsymbol{\xi}(t_k)\right)\right], \tag{2}$$

where $\boldsymbol{\Sigma}$ defines the Gaussian widths in CV space. In conventional metadynamics, once the relevant CV space has been sampled sufficiently, the accumulated bias approaches the negative of the underlying free-energy surface, up to an additive constant,

$$F(\boldsymbol{\xi}) \simeq -V_{\text{bias}}(\boldsymbol{\xi}, t) + C. \tag{3}$$

To limit continued growth of the bias, and to obtain a controlled asymptotic distribution, well-tempered metadynamics has been developed.[33] In this formulation, the height of each newly deposited Gaussian decreases according to the bias already accumulated at this CV

value,

$$w_k = w_0 \exp\left[-\frac{V_{\text{bias}}\left(\boldsymbol{\xi}(t_k), t_k\right)}{k_{\text{B}}\Delta T}\right], \tag{4}$$

such that

$$V_{\text{bias}}(\boldsymbol{\xi}, t) = \sum_{t_k<t} w_k \exp\left[-\frac{1}{2}\left(\boldsymbol{\xi} - \boldsymbol{\xi}(t_k)\right)^{\text{T}} \boldsymbol{\Sigma}^{-1} \left(\boldsymbol{\xi} - \boldsymbol{\xi}(t_k)\right)\right]. \tag{5}$$

Here, ΔT controls the rate at which bias deposition is tempered. It is commonly expressed through the bias factor

$$\gamma = \frac{T + \Delta T}{T}, \tag{6}$$

where T is the physical simulation temperature. In the long-time limit, the well-tempered bias is related to the free-energy surface by

$$V_{\text{bias}}(\boldsymbol{\xi}, t \to \infty) = -\frac{\Delta T}{T + \Delta T} F(\boldsymbol{\xi}) + C, \tag{7}$$

or, equivalently,

$$F(\boldsymbol{\xi}) = -\frac{\gamma}{\gamma - 1} V_{\text{bias}}(\boldsymbol{\xi}, t \to \infty) + C. \tag{8}$$

Thus, unlike conventional metadynamics, the accumulated well-tempered bias does not completely compensate the underlying free-energy surface. Instead, the progressively decreasing hill heights provide controlled convergence while reducing overfilling of free-energy minima.[33]

Umbrella Sampling

Umbrella sampling uses harmonic bias potentials along a collective variable that restrain the system to selected regions of CV space.[34,35,37] For window i, centered at ξ_i, the harmonic bias potential is defined as

$$V_{\text{bias}}^{(i)}(\xi) = \frac{\kappa_i}{2}\left(\xi - \xi_i\right)^2, \tag{9}$$

where κ_i is the corresponding force constant. Independent simulations are performed for a series of window centers ξ_i, chosen such that neighboring windows sample overlapping

regions of the CV. The resulting biased distributions are then combined to reconstruct the underlying free-energy profile.[37] In the present work, the reconstruction is performed using umbrella integration.[36] For each umbrella window, the derivative of the free energy can be estimated from the displacement of the mean collective variable, $\bar{\xi}_i$, from the umbrella center ξ_i, provided that the biased distribution is approximately unimodal and symmetric,

$$\left.\frac{\partial F}{\partial \xi}\right|_{\bar{\xi}_i} = -\kappa_i \left(\bar{\xi}_i - \xi_i\right). \tag{10}$$

These derivative estimates can be interpolated, e.g. using Gaussian Process Regression, and integrated to reconstruct the free-energy profile.[43]

Free-Energy Differences

Both Metadynamics and Umbrella Sampling assume that the sampling along the slow degrees of freedom can be accelerated using biases defined in the CVs. The resulting FES thus contains the entropy of all degrees of freedom, except for the ones described by the CVs. The relative stability of two states can be obtained by integrating the reconstructed free-energy surface over their corresponding basins. This is, in general, not equal to the difference between the values of the free-energy surface at the respective minima. In particular, differences in the width and shape of the basins contribute to the relative populations and therefore to the free-energy difference.[44]

For a state A, occupying a region Ω_A of the CV space, its statistical weight is proportional to

$$P_A \propto \int_{\Omega_A} \exp\left[-\beta F(\boldsymbol{\xi})\right] \mathrm{d}\boldsymbol{\xi}, \tag{11}$$

where $F(\boldsymbol{\xi})$ is the free-energy surface and

$$\beta = \frac{1}{k_\mathrm{B} T}. \tag{12}$$

The corresponding basin free energy can therefore be written, up to an additive constant, as

$$F_A = -k_\mathrm{B}T \ln \left[\int_{\Omega_A} \exp\left[-\beta F(\boldsymbol{\xi})\right] \, \mathrm{d}\boldsymbol{\xi} \right] + C. \tag{13}$$

For two states, the free-energy difference is consequently

$$\Delta F_{\mathrm{A-B}} = F_\mathrm{A} - F_\mathrm{B}, \tag{14}$$

which can be evaluated directly from the integrated statistical weights,

$$\Delta F_{\mathrm{A-B}} = -k_\mathrm{B}T \ln \left[\frac{\int_{\Omega_\mathrm{A}} \exp\left[-\beta F(\boldsymbol{\xi})\right] \, \mathrm{d}\boldsymbol{\xi}}{\int_{\Omega_\mathrm{B}} \exp\left[-\beta F(\boldsymbol{\xi})\right] \, \mathrm{d}\boldsymbol{\xi}} \right]. \tag{15}$$

Methods

biASE

biASE is implemented as a modular biasing layer on top of atomistic simulation workflows based on ASE. The biasing potential is defined in CV space, whereas the CV is defined in Cartesian coordinates. Both may be arbitrary mathematical functions that can be expressed as differentiable Python functions in JAX.[45] Here, JAX is used for automatic differentiation of the CVs with respect to the Cartesian coordinates and of the bias potential with respect to the CV(s). To obtain the bias forces in Cartesian space, the chain rule is applied; Figure 1 shows the full schematic workflow of biASE. The resulting bias forces are then added to the forces from the underlying potential, which allows the same framework to be used with conventional electronic-structure codes, MLIPs, and classical FFs, as long as they provide an ASE interface.

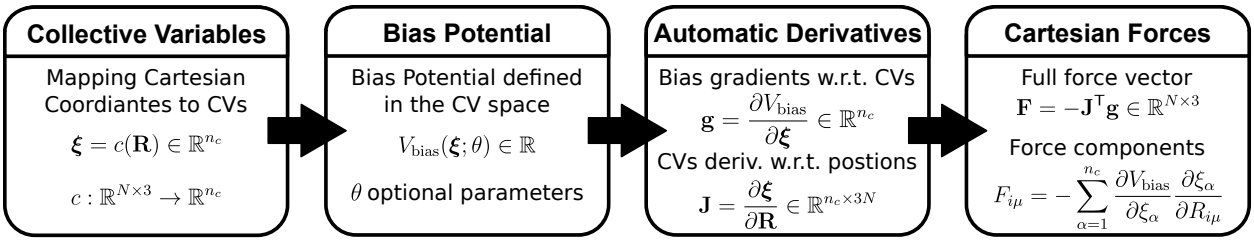


Figure 1: Schematic workflow of biASE, starting from the definition of the CV in Cartesian space, followed by the definition of the bias potential in CV space, and finally using automatic differentiation and the chain rule to obtain the corresponding forces in Cartesian space.

Model Overview

A comprehensive benchmark of molecular foundation models is beyond the scope of this work, and fundamentally hard to achieve given the speed of the development in the field. Instead, we aim to investigate some selected models in order to draw conclusions on the influence of model architecture, training set size, and long-range interactions.

MACE-OFF24(M) is a $E(3)$-equivariant MACE model with two interaction layers, a 6 Å cutoff, and 1.4 M parameters.[11,25] It was trained on approximately 1.2 million charge-neutral configurations from the SPICE.v2 dataset, computed at the ωB97M-D3(BJ)/def2-TZVPPD level of theory and covering ten elements (H, C, N, O, F, P, S, Cl, Br, I).[25,46,47] Thhis model is intended to predict energies and forces for neutral organic systems, including diverse molecular conformers, noncovalent complexes, peptides, solvated amino-acid environments, water clusters, and protein–ligand-like configurations.

PET-SPICE is based on the Point Edge Transformer (PET) architecture. In contrast to MACE, PET does not impose rotational equivariance; instead, the symmetry is learned from rotationally augmented training data, while permutation and translation invariance are retained exactly.[22] The model uses three interaction layers, a 4.5 Å cutoff, and 190 M parameters. It was trained on the curated SPICE.v2 data and official splits introduced with MACE-OFF, i.e. on neutral molecules at the ωB97M-D3(BJ)/def2-TZVPPD level.[22,25,46]

MACE-MH-1 is a multi-head MACE foundation model that combines a shared representation with separate output heads associated with different electronic-structure data

domains.[19] It consists out of a shared backbone with two interaction layers and a 6 Å cutoff is combined with separate shallow readout heads for different levels of electronic-structure theory. All heads included MACE-MH-1 has 6.4 M parameters. The backbone was pre-trained on 100 million inorganic-crystal configurations from OMat24 at the PBE/PBE+U level, spanning 89 elements.[17] The OMol head used herein was subsequently fine-tuned on 1 % of the OMol25 dataset computed at the ωB97M-V/def2-TZVPD level. This subsampling results in 1.2 million neutral, closed-shell and diverse organic, organometallic and transition-metal configurations.Levine et al.[18], Batatia et al.[19] Its architecture makes the model a single potential across materials, surface, and molecular chemistry, with the OMol head recommended for molecular systems.

MACELES-OFF combines a short-range MACE model with Latent Ewald Summation (LES) for long-range electrostatic interactions.[26,27] It has two interaction layers, a 4.5 Å cut-off, 1.9 M parameters. LES predicts latent atomic charges from local features and evaluates their long-range electrostatic interaction by Ewald summation, without requiring reference charges during training. The model was trained on the same neutral SPICE.v1 data split as MACE-OFF23 (approximately 0.95 million configurations) at the ωB97M-D3(BJ)/def2-TZVPPD level.[25,27] Its intended domain matches that of MACE-OFF, which makes it a good comparison point to isolate the effect of explicit long-range electrostatics.

MACE-POLAR-1-M augments a local MACE backbone with an explicit description of nonlocal electrostatics and electrostatic induction.[28] It contains two interaction layers with a 6 Å cutoff and 1.5 million parameters. Rather than assigning point charges, MACE-POLAR learns atom-centered, spin-resolved charge-density multipoles. This provides an environment-dependent representation of polarization, charge redistribution, and long-range electrostatic interactions while retaining the many-body MACE description of short-range chemistry.[28] MACE-POLAR-1 was trained on approximately 100 million molecular configurations of OMol25 at the ωB97M-V/def2-TZVPD level, covering a broad range of molecular compositions, charge and spin states, conformers, complexes, and solvated environments.[18,28]

The model is intended for general molecular chemistry, including charged and open-shell species, ions in solution, transition-metal complexes, and protein–ligand systems.

SO3LR combines the $SO(3)$-equivariant SO3krates network with physics-based pairwise terms for short-range repulsion, long-range electrostatics, and dispersion.[29] The model has three interaction layers, a 4.5 Å cutoff and 0.53 M parameters. Electrostatics are evaluated from machine-learned partial charges, and dispersion from a universal pairwise van der Waals potential parameterized by machine-learned Hirshfeld ratios. The model was trained on approximately 4.0M neutral and charged molecular configurations computed at the PBE0+MBD level, covering eight elements (H, C, N, O, F, P, S, Cl).[29] Since SO3LR was trained on forces, dipole moments, and Hirshfeld ratios but not on absolute energies, only relative energies are meaningful. The model is intended for biomolecular simulations in explicit solvent, including peptides, proteins, carbohydrates, and lipids.

Enhanced Sampling Simulations

All simulations were performed with ASE and the biASE framework for differentiable collective variables and bias potentials. Unless stated otherwise, the equations of motion were integrated with a Langevin thermostat at $T = 293$ K, using a friction coefficient of 0.01 fs^{-1} and a timestep of 0.5 fs. Energies and forces were supplied either by the respective MLIP, by AMBER force fields through OpenMM, or by the ωB97M-V/def2-TZVPD reference calculations, using ORCA 6.1.1. Free-energy profiles are reported relative to their respective minima. The test systems and their respective CVs are shown in Figure 2.

Malondialdehyde and Fluoromalondialdehyde

Malondialdehyde is a well-known model system for intermolecular proton transfer, making it an ideal candidate for evaluating molecular foundation models.[48] We additionally consider the substituted Fluoromalondialdehyde, which has a substantially distorted FES, compared

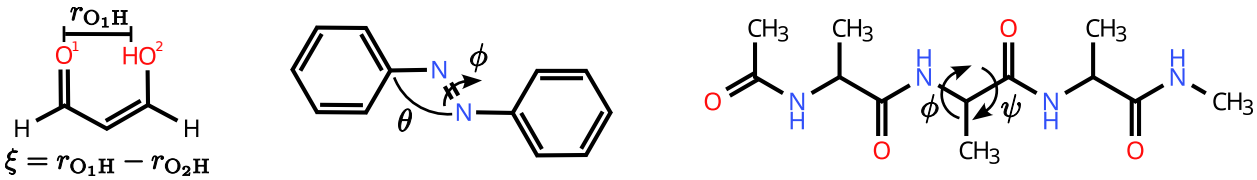


Figure 2: Collective variables used for enhanced-sampling simulations. Left: O-H distance used to construct the proton-transfer coordinate in malondialdehyde and fluoromalondialdehyde. Center: CNN bending angle θ and CNNC torsional angle ϕ used to describe azobenzene isomerization. Right: backbone dihedral angles ϕ and ψ of the central alanine residue in Ac-Ala$_3$-NMe.

to the unsubstituted case. Due to the small size of these systems, reference free-energy simulations at the ωB97M-V/def2-TZVPD level of theory could be performed.

The FESs for proton transfer in malondialdehyde and fluoromalondialdehyde were obtained via Umbrella Sampling along the proton-transfer coordinate

$$\xi = r_{O_1H} - r_{O_2H}, \tag{16}$$

where r_{O_1H} and r_{O_2H} are the distances between the transferring hydrogen atom and the two oxygen atoms. A harmonic umbrella potential,

$$V_{\text{umb}}(\xi) = \frac{\kappa}{2}(\xi - \xi_0)^2, \tag{17}$$

with $\kappa = 50$ eV Å^{-2} was used. Window centers were spaced by 0.05 Å. Each window was initialized by constrained optimization of the biased potential, followed by 10 ps of Langevin dynamics. For the free-energy analysis via Umbrella Integration, the first 1 ps of dynamics was discarded for equilibration.

Azobenzene

Azobenzene is a well-studied photoswitch that exploits *cis–trans* isomerization.[49] The ground-state free energy barrier for this process is of interest because it determines the rate of the thermal backreaction, and therefore the half-life of the switched state.[10.1021/acs.jpca.1c01695,

10.1002/anie.201506126] It should be noted that a triplet mediated mechanism has been reported to play an important role here.[10.1021/acscentsci.2c00897] In the present work, we focus on the singlet ground state FES, however, since currently most molecular foundation models are not able to treat different spin states and none include spin-orbit-coupling. To describe the corresponding FES, two-dimensional well-tempered metadynamics was used,[32,33] with the CNN bending angle, θ, and the CNNC torsional angle, ϕ as CVs. Gaussian bias potentials of initial height 1 kJ mol^{-1} and widths $\sigma_\theta = \sigma_\phi = 10^\circ$ were deposited every 100 MD steps. Simulations used a timestep of 0.5 fs, $T = 293$ K, and a well-tempered bias factor of $\gamma = 10$.

Due to the higher complexity of this system, direct DFT free energy calculations were computationally prohibitive as a reference. Instead, a dataset of 2,400 configurations was assembled by randomly sampling 200 snapshots from 12 independent metadynamics runs using MACE-OFF24, MACE-OMOL, MACE-MH-1, MACE-POLAR, PET-SPICE, and SO3LR. This data was randomly split into training, validation, and test sets (in a 80:10:10 ratio) and used to train a MACE model, with a cutoff of 6.0 Å, two interaction blocks, $128\times0e+128\times1o$ hidden irreducible representations, and correlation order 3. Training was performed for up to 300 epochs using an initial energy/force weighting of 100/10, followed by 200 epochs with weights of 10/10. The Supporting Information section *Training Data - Azobenzene* shows the datasets mapped onto the CV spaces.

As additional references, two of the foundation models were fine-tuned on this data. Specifically, MACE-MH-1 was fine-tuned from the OMOL foundation head using multi-head fine-tuning with energy and force weights of 1 eV^{-1} and 100 (eV/Å)$^{-1}$, respectively, a learning rate of 10^{-3}, and float64 precision for 200 epochs.For SO3LR, the default pretrained potential was fine-tuned using AdamW, a learning rate of 10^{-4} with exponential decay, and force and energy loss weights of 1 and 0.01, respectively, for 200 epochs. Long-range interactions were enabled with a 1000 Å cutoff.

In all cases, independent simulations were initiated from optimized *cis* and *trans* struc-

tures. Smooth angular restraining walls (starting at $\theta = 170°$) were applied to prevent excursions into unphysical regions of CV space while retaining continuous bias forces. The final production trajectories were extended to 3 ns per starting isomer. Free-energy surfaces were reconstructed from the deposited Gaussian bias, shifted to a zero minimum, and averaged across the *cis*- and *trans*-initiated simulations.

Transition pathways were approximated using a minimax criterion: among smooth paths connecting the *cis* and *trans* basins, the path that minimizes the highest free energy encountered along the path was selected. The low-θ route was constrained to the $\theta = 105$–$135°$ corridor, whereas the high-θ route was constrained to pass through the lowest free-energy point within $\theta = 155$–$175°$.

Solvated Trialanine

The trialanine peptide is commonly used as a simple prototype for a biomolecule in solution. As such, metadynamics simulations of this system, using the central backbone dihedral angles ϕ and ψ as CVs have previously been used as a benchmark for molecular foundation models.[25] However, defining an adequate reference is challenging in this case, since ab initio MD is prohibitively expensive for systems of this size, and even single point calculations with the range-separated hybrid functionals used for training the MLIPs are not possible. Consequently, the MLIP simulations are usually compared to simulations with classical forcefields, which have been calibrated for describing solvated bio-molecules. This comparison is meaningful in terms of the location and relative stability of different folded states, but quantitative agreement of the FESs should not be expected. Herein we use AMBER ff14SB/TIP3P[50,51] and AMBER ff19SB/OPC[52,53] reference surfaces, obtained from four independent 1 ns metadynamics runs each. The Supporting Information, also includes a comparison of our biASE calculations with the AMBER ff14SB/TIP3P OpenMM reference using OpenMM's native metadynamics implementation.

The solvated tripeptide Ac–Ala_3–NMe simulation cell was constructed with the Amber-

Tools `tleap` program and parameterized with AMBER ff14SB and TIP3P water. The peptide was placed in an approximately 31 Å cubic solvent box containing 631 water molecules. Before metadynamics, the system was minimized, heated from 0 to 300 K for 30 ps in the NVT ensemble, and equilibrated for 100 ps in NPT at 293 K and 1 bar using isotropic Berendsen pressure coupling with a 2 ps coupling time and 1 fs time steps, using the classical force field. This was followed by 20 ps of dynamics in the NVT ensemble at 293 K and 100 ps in NPT ensemble at 293 K and 1 bar using the respective model with 0.5 fs time steps. The FESs were then obtained from 1 ns simulations using two-dimensional well-tempered metadynamics in the backbone dihedral angles ϕ and ψ of the central alanine residue. Gaussian hills of height 1 kJ mol^{-1} and widths $\sigma_\phi = \sigma_\psi = 15^\circ$ were deposited every 100 MD steps. All simulations used $T = 293$ K, a timestep of 0.5 fs, and a well-tempered bias factor of 10.[32,33] These settings are in line with what has been used for benchmarking foundation models previously. In all cases, free-energy surfaces were reconstructed from the accumulated bias, shifted such that their minimum was zero, and displayed on a common energy scale.

Results and Discussion

Malondialdehyde

The FESs for malondialdehyde and fluoromalondialdehyde obtained with the ωB97M-V/def2-TZVPD reference and the different MLIPs are shown in Figure 3. This comparison shows that the most recent models (here MACE-MH-1 and MACE-POLAR-1-M) are closest to the DFT reference, differing from it by roughly 1 kJ mol^{-1}. This is a clear indication of the quality of the models and datasets for predicting non-equilibrium properties of organic gas-phase molecules. These models have the advantage of being trained at exactly the same level of theory as the reference, whereas the SPICE-based fMLIPs, PET-SPICE and MACE-OFF24(M), are trained on ωB97M-D3(BJ)/def2-TZVPPD and significantly overpredict the energy barrier, with MACE-OFF24(M) performing notably worse than PET-SPICE in this respect. On the other hand, all equivariant models correctly predict the free energy surface to be symmetrical, while PET-SPICE shows a significant discrepancy between the two equivalent minima. This shows that lack of equivariance can lead to subtle errors in advanced simulation workflows. SO3LR underestimates the barrier to a similar degree as MACE-OFF24(M) overestimates it. Since SO3LR is trained on PBE0+MBD reference data, this difference could be due to a discrepancy between the reference levels of theory.

When one of the aldehyde groups is fluorinated, the FES becomes highly asymmetric. Qualitatively, all methods show that the O-H bond is strongly destabilized on the carbon bearing the fluorinedue to the electron-withdrawing character of fluorine. For fluoromalondialdehyde, MACE-MH-1 and MACE-POLAR-1-M again show agreement within chemical accuracy with the reference method for the energy barrier and relative stability of the two tautomers. PET-SPICE is quite close as well,showing errors of at most 2 kJ mol^{-1}. In contrast, MACE-OFF24(M) significantly overstabilizes tautomer A, with an error of roughly 9.5 kJ mol^{-1}. SO3LR also underestimates the free-energy difference significantly, and consequently underestimates the energy barrier for proton transfer from tautomer B to A, while

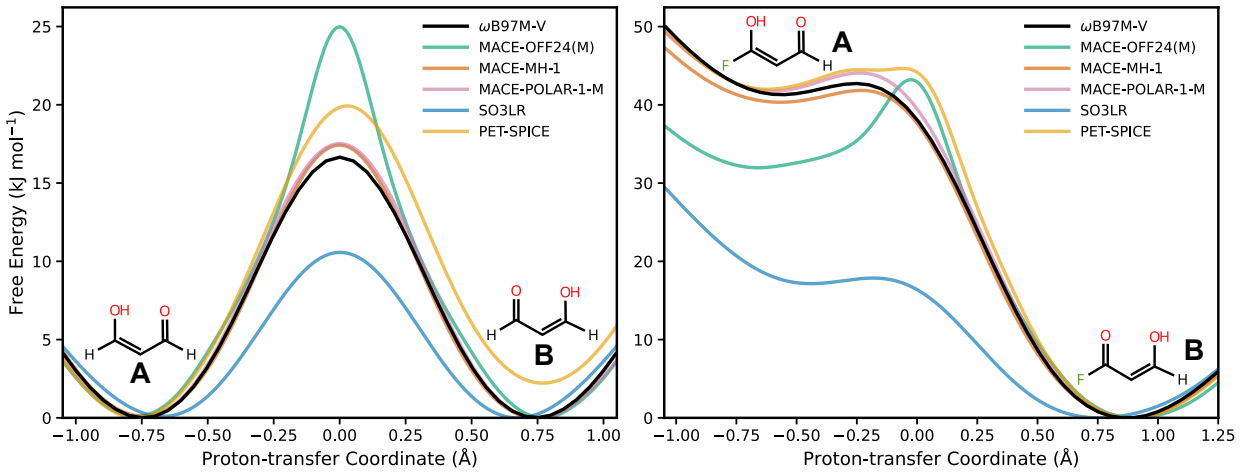


Figure 3: Free-energy profiles from umbrella sampling at 293 K followed by umbrella integration for the hydrogen transfer of malondialdehyde (left) and fluoromalondialdehyde (right), calculated with different MLIP models and the ωB97M-V reference shown in black.

the barrier from A to B is quite accurate, with an error of 0.7 kJ mol^{-1}. Again, the most likely explanation for this behaviour is the different reference level of theory. Table 1 shows the energy barriers for malondialdehyde and fluoromalondialdehyde as well as the stabilization energy of the two fluoromalondialdehyde tautomers.

Table 1: Free-energy barriers and stabilization energies for proton transfer in malondialdehyde and fluoromalondialdehyde at 293 K, in kJ mol^{-1}. $\Delta F^{\ddagger}_{\text{malon}}$ denotes the energy barrier for malondialdehyde. $\Delta F^{\ddagger}_{A\to B}$ and $\Delta F^{\ddagger}_{B\to A}$ are the directional barriers for the two tautomers A and B, and $\Delta F_{A-B} = F_A - F_B$ is the free-energy difference between the two fluoromalondialdehyde tautomers.

Model	$\Delta F^{\ddagger}_{\text{malon}}$	$\Delta F^{\ddagger}_{A\to B}$	$\Delta F^{\ddagger}_{B\to A}$	ΔF_{A-B}
ωB97M-V	16.6	1.4	42.7	40.9
MACE-OFF24(M)	24.9	11.3	43.2	31.4
MACE-MH-1	17.4	1.5	41.8	39.7
MACE-POLAR-1-M	17.5	2.3	44.1	41.4
SO3LR	10.5	0.7	17.9	17.4
PET-SPICE	18.8	2.7	44.7	41.2

Azobenzene

Figure 4 shows the FES for the thermal isomerization of azobenzene, obtained via well-tempered metadynamics. Additionally, the free energies of isomerization $\Delta F_{\text{trans}-\text{cis}}$ and

reaction barriers $\Delta F^{\ddagger}$ are shown in Table 2. Here, the MACE model trained from scratch on configurations from all foundation model metadynamics runs serves as a reference. It predicts a stabilization of the trans isomer by 47 kJ/mol relative to the cis isomer and finds two reaction pathways via possible transition states at CNN angles of roughly $\theta = 120°$ and $\theta = 160°$, both with a dihedral angle around $\phi = 90°$. Both of these pathways have similar free energy barriers of around 130-140 kJ/mol. For reference, the isomerization energy was previously reported to be 64 kJ/mol at the B3LYP/6-31G* level,[49] and 55 kJ/mol at the B3LYP-D3/6-311G(d,p) CITE[10.1002/anie.201506126]. The latter study also reported cis-to-trans isomerization energy barriers of 101 and 165 kJ/mol for B3LYP/6-311G(d,p) with and without the D3 correction, respectively. This indicates that differences in functional, dispersion correction, and basis set will have a substantial influence on the relative stability of different states for this system, independent of the free energy contributions.

Except for SO3LR, where the FES only has a single minimum at the *trans* position, the foundation models give qualitatively similar results. The *cis* minimum is typically located around ($\phi = 0°$, $\theta = 120°$), and the *trans* minimum around ($\phi = \pm 180°$, $\theta = 115°$). The respective stabilization energies are between 48.1 and 43.2 kJ mol^{-1} for the MACE models and PET-SPICE, and thus within chemical accuracy of the values of the bespoke MACE model. For the transition pathways between the minima, larger differences are observed. Here, MACE-MH-1 shows the largest deviation from the reference and does not reproduce the $\theta = 160°$ isomerization pathway at all. In contrast, MACE-POLAR-1-M reproduces the reference FES to within the sampling accuracy of the metadynamics simulations. Since the incorporation of long-range electrostatics is unlikely to be significant for this relatively small system, this suggests that the substantially larger amount of molecular training data in the OMOL dataset improves the accuracy of the model also in regions far from equilibrium. Nevertheless, the MACE-OFF24(M) and PET-SPICE models also perform quite well, though they underestimate the free-energy reaction barriers by 15-30 kJ/mol.

Arguably, the zero-shot prediction of FESs is not a necessary condition for a foundation

model to be useful. At least, however, it should be possible to obtain accurate results when fine-tuning the models. To explore this, fine-tuning was performed for the SO3LR and MH1 models, using the same training set as the bespoke MACE model. For the former, the resulting FES recovers both pathways and energetics close to the reference. The fine-tuned SO3LR model also improves strongly. However, despite the relatively large fine-tuning dataset, significant discrepancies with the reference MACE model (and all other models) remain. Most notably, the FES only displays a single transition pathway around $\theta = 160°$. This likely reflects one of the core design choices of SO3LR, namely the use of physics-based pairwise contributions for repulsion, dispersion, and electrostatics. These are, on one hand, a useful inductive bias, which can help the model be robust in the low-data regime. On the other hand, they may also limit the flexibility of the model in a fine-tuning setting. Alternatively, this discrepancy may also simply be the result of negative transfer, since the SO3LR foundation model is a poor starting point in this particular case.

Overall, we find that a frontier model such as MACE-POLAR-1-M can be as accurate as a bespoke MLIP, even for a challenging 2D metadynamics simulation. Models trained on smaller datasets, such as MACE-MH-1 do not reach quantitative agreement with the reference method. Nevertheless, the discrepancy between most tested MLIPs (with the exception of SO3LR) is comparable to the difference between different DFT functionals, and significantly smaller than the errors of semiempirical methods such as the DFTB3 approach reported in Ref. 49.

Table 2: Free energies of reaction $\Delta F_{\mathrm{trans-cis}}$ and free energy barriers $\Delta F^{\ddagger}$ for cis-to-trans isomerization of azobenzene from well-tempered metadynamics at 293 K. Reported values are averages over the *cis*- and *trans*-initiated reconstructions, with the uncertainty given by the standard deviation. The SO3LR foundation model is not shown because it lacks a stable *cis* state. All values are given in kJ mol^{-1}.

Model	$\Delta F^{\ddagger}_{120^\circ}$	$\Delta F^{\ddagger}_{160^\circ}$	$\Delta\Delta F^{\ddagger}_{\mathrm{TS}}$	$\Delta F_{\mathrm{trans-cis}}$
MACE-OFF24(M)	113.9 ± 6.5	116.4 ± 1.1	2.5 ± 7.0	-45.4 ± 1.8
MACE-MH-1	92.4 ± 1.6	111.0 ± 0.7	18.6 ± 1.8	-43.2 ± 0.9
MACE-POLAR-1-M	128.2 ± 0.9	132.2 ± 2.2	4.0 ± 1.6	-48.1 ± 0.4
PET-SPICE	113.0 ± 8.0	105.3 ± 2.2	-7.7 ± 9.3	-47.8 ± 0.7
MACE from scratch	135.0 ± 7.3	133.6 ± 1.5	-1.4 ± 5.8	-46.6 ± 3.7
Fine-tuned SO3LR	165.4 ± 12.3	150.4 ± 6.1	-15.0 ± 18.1	-61.7 ± 2.8
Fine-tuned MACE-MH-1	128.8 ± 9.7	127.7 ± 2.0	-1.1 ± 11.3	-49.1 ± 3.2

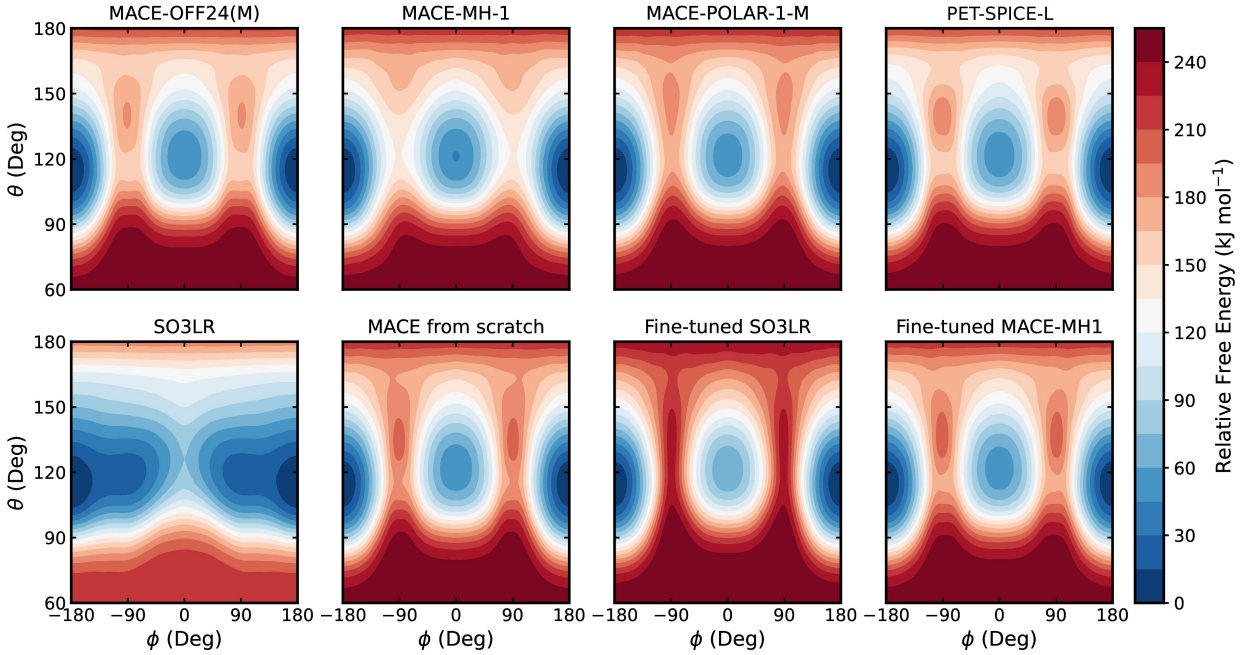


Figure 4: Symmetrized free-energy surfaces of azobenzene as functions of the CNNC dihedral angle, ϕ, and the selected CNN bond angle, θ, for different MLIPs from well-tempered metadynamics at 293 K. The *cis*- and *trans*-initiated surfaces were averaged pointwise, and the resulting surface was symmetrized with respect to dihedral inversion, $F(\phi, \theta) = F(-\phi, \theta)$.

Trialanine in Water

The previous two examples are intended to investigate how robustly different foundation models behave in non-equilibrium regions of the potential energy surface, which are generally less well represented in the training sets. The systems themselves, however, were small organic molecules in vacuum, which are clearly within the domain of applicability of molecular foundation models. However, many (most) scientific questions in chemistry and materials science concern systems in the condensed phase. The transfer from isolated training systems to the condensed phase therefore represents an important challenge. To address this, all investigated foundation models were trained on intermolecular dimers and clusters, and transferability to molecular liquids, crystals and solvated bio-molecules has been demonstrated to some extent. However, striking the delicate balance between inter- and intramolecular forces in the condensed phase from cluster data alone is extremely challenging, so that quantitative accuracy is usually not achieved.[0.1021/acs.jctc.6c00735] In this context, models that explicitly model long-range electrostatics are of great interest, since they provide a physically reasonable Madelung potentials for condensed phase systems, even when only trained on isolated systems.[CITE Martin JCTC, POLAR, LES]

Figure 5 shows the predicted FESs of the Ac–Ala_3–NMe trialanine peptide in explicit aqueous solution. Two classical biomolecular forcefields are included for reference, namely AMBER ff14SB and ff19SB using the TIP3P and OPC water models, respectively.[CITE!] Unlike MLIPs, classical force fields (and the corresponding water models) are not only fitted to extensive quantum mechanical reference data, but also calibrated against experimental data to reproduce known secondary structure elements in peptides and experimental properties of water. Perfect agreement between classical forcefields across the FES is therefore not to be expected (or even desirable). Instead, the focus should be mainly on the location and relative stability of the minima on the FES. Here, both reference methods agree on four main states, corresponding to a β-sheet-like basin (aβ), a polyproline-II-like basin (PPII), a right-handed α-helical basin αH(R), and a left-handed α-helical basin αH(L).[50,54,55] Among

these, both forcefields predict the PPII basin to be the thermodynamic minimum, followed by the aβ, which lies ca. x kJ/mol higher in energy. Finally, the two α-helical basins lie higher in energy by ca. x kJ/mol. Perhaps surprisingly, the left- and right-handed basins are almost identical in energy, particularly at the ff19SB/OPC level of theory. The detailed energy differences between each minimum and the lowest minimum for all models are shown in Table 3. For the classical forcefields, this also includes as error estimates the standard deviations of four independent 1 ns metadynamics simulations. This highlights that such short runs can display significant variability, on the order of x kJ/mol in relative energies. While it would of course be possible to run the classical force-fileds for longer simulation times, this becomes computationally prohibitive for MLIPs. As a consequence, 1 ns metadynamics runs have been the standard for reported peptide MLIP FESs.[CITE] To obtain an overview over multiple MLIPs we follow this convention, but emphasize that small energy differences should not be overinterpreted.

The most apparent feature when comparing the FESs is that none of the foundation MLIPs reproduces the pronounced energy barrier observed for AMBER ff14SB/TIP3P around $\phi = 120°$. Such a pronounced barrier is also not observed for AMBER ff19SB/OPC or in DFT calculations with implicit solvent for alanine dimers and trimers.[50,54] In ff19SB this barrier was removed by introducing a coupling term between the dihedral angles in each residue. The many-body nature of modern MLIPs naturally captures this effect. At the same time, the three maxima at $\phi = 0°$ are reproduced by all methods.

Concerning the minima on the FES, most models predict low-energy states consistent with the PPII, αH(R), and αH(L) basins. In contrast, reproducing a minimum corresponding to the aβ basin is more challenging. For several models, such as PET-SPICE or MACE-POLAR-1-M, this minimum appears to merge with the PPII basin. The SPICE-based MLIPs all predict PPII to be the lowest-energy conformer, whereas MACE-POLAR-1-M slightly overstabilizes αH(R), MACE-MH-1 aβ, and SO3LR αH(L). However, these deviations are smaller than the sampling accuracy of the simulations, so that they should not be over-

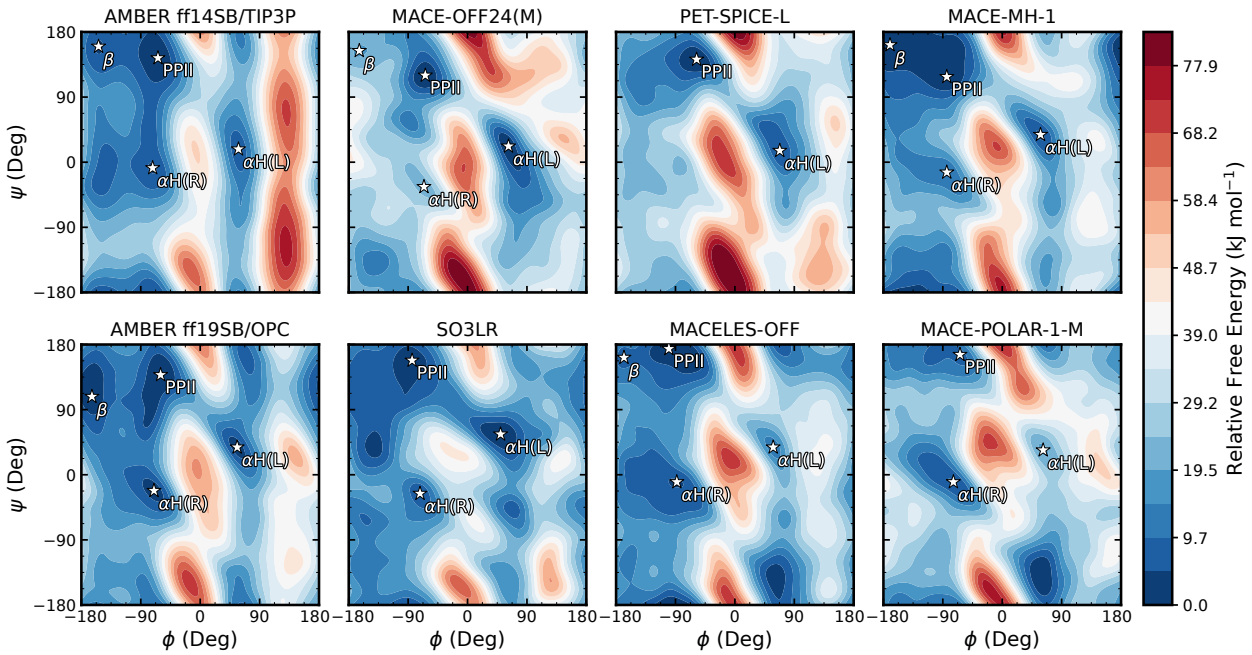


Figure 5: Free-energy surfaces of solvated Ac–Ala_3–NMe from well-tempered metadynamics at 293 K using AMBER ff14SB/TIP3P, AMBER ff19SB/OPC, and molecular foundation models. The collective variables are the backbone dihedral angles ϕ and ψ of the central alanine residue. The AMBER reference surfaces are averages over four independent 1 ns trajectories; all other surfaces are based on 1 ns simulations. Minima were selected within the constrained search regions.

Table 3: Point free-energy differences (kcal mol^{-1}) between highlighted solvated Ac–Ala$_3$–NMe conformational basins from well-tempered metadynamics at 293 K. Minima were selected within the constrained search regions and values are referenced to the lowest labeled minimum for each model (bold). AMBER values are means from four independent 1 ns simulations, with standard deviations from the PPII-referenced replica values; MLIP values are from single 1 ns simulations. A dash denotes a basin for which no eligible local minimum was identified.

Model	aβ	PPII	αH(R)	αH(L)
AMBER ff14SB/TIP3P	0.9 ± 1.3	**0.0**	2.0 ± 0.8	1.6 ± 2.5
AMBER ff19SB/OPC	0.3 ± 2.3	**0.0**	1.5 ± 1.8	1.5 ± 2.1
MACE-OFF24(M)	4.5	**0.0**	5.3	0.1
PET-SPICE	–	**0.0**	–	1.4
MACE-MH-1	**0.0**	0.1	2.7	1.0
SO3LR	–	0.5	1.5	**0.0**
MACELES-OFF	2.0	**0.0**	1.1	2.9
MACE-POLAR-1-M	–	0.9	**0.0**	3.3

interpreted. Similarly, the relative stability of αH(R) and αH(L) is highly inconsistent between different models, with MACE-OFF24(M) predicting αH(L) to be nearly isoenergetic with the ground state and MACE-POLAR-1-M finding αH(R) to be more stable.

As mentioned above, the application to condensed phase simulations is one of the key motivations for the development of long-ranged MLIPs, such as SO3LR, MACELES-OFF, and MACE-POLAR. In this light, it is interesting to note that these models (each trained on a different training set) indeed share some similarities. In particular, the free energy maxima around $\phi = 0°$ are generally less repulsive than for the short-ranged MLIPs, and thus more similar to the reference force-fields. Both long-ranged MACE models also predict an additional minimum around $\psi = -165°$, $\phi = 75°$, which is much less pronounced in either of the AMBER force-field surfaces.[27] It is generally remarkable that the MACELES-OFF and MACE-POLAR-1-M surfaces are more similar than the MACELES-OFF and MACE-OFF24(M) surfaces. This indicates that the inclusion of long-range interactions is more significant than the size and nature of the training set in this case.

Overall, the sensitivity in the ordering of the minima highlights that the trialanine FES remains a challenging test for transferable molecular foundation models. Current foundation

models, such as MACE-POLAR-1-M, MACELES-OFF, and MACE-MH-1, already qualitatively reproduce the reference surfaces and reproduce the positions of the main minima reasonably well. This should, however, be interpreted in light of the approximately 8 kJ mol^{-1} uncertainty associated with well-tempered metadynamics sampling over the relatively short simulation time of 1 ns. In other words, these models are now good enough that more rigorous statistical convergence of enhanced sampling simulations is required to identify potential shortcomings. A more quantitative assessment will require multiple independent metadynamics runs and longer simulation times to improve the statistical precision of the predicted FESs.

Conclusion

In this work, different molecular foundation models were evaluated across a range of test cases for free-energy sampling, providing an extensive assessment of their performance outside the near-equilibrium regime. The results demonstrate that the most recent foundation models, trained on datasets containing more than 100 million systems, are capable of reproducing free-energy profiles for organic gas-phase systems, such as malondialdehyde and fluorinated malondialdehyde. However, small energy differences between competing pathways can still lead to qualitatively incorrect predictions of reaction mechanisms, as observed in the torsional analysis of azobenzene, where even models with extensive training datasets, such as MACE-MH-1, miss relevant reaction pathways. For solvated systems, the results also demonstrate the importance of long-range electrostatic interactions, which have an influence on the FES that is comparable to the network architecture or training set composition. Overall, this free-energy analysis shows that the most recent molecular foundation models are already highly capable of reproducing complex free-energy surfaces. When combined with electrostatic corrections, they can achieve accuracies close to the uncertainty of enhanced sampling methods, such as well-tempered metadynamics, within reasonable sampling times. While molecular foundation models improved through larger datasets and additional long-range electrostatics are promising for free-energy sampling, they remain at least one step away from being truly general or universal models for molecular systems, particularly in regions far from equilibrium.

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Code & Data Availability

The biASE code can be found at `https://github.com/Nilsgoe/biASE`. The benchmarks can be found at `https://github.com/Nilsgoe/Enhanced_Sampling`.

Author Declarations

The authors have no conflicts to disclose.

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